



A new method to establish NMR T_2 spectrum based on bimodal Gaussian density function: A case study of tight sandstone in East China Sea Basin

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ABSTRACT

The identification and evaluation of tight sandstone reservoirs has been very difficult due to their low porosity, low permeability, complex pore structure and strong heterogeneity. In order to evaluate the quality of tight sandstone reservoir, priority should be given to the quantitative evaluation of pore size. Nuclear magnetic resonance (NMR) log is frequently employed in pore size evaluation; however, due to its high technical costs and other limitations, it can only be conducted in very few wells and the wide application can never be found. In this study, 12 core samples from the tight sandstone in East China Sea Basin have been collected to conduct centrifugal NMR experiments. Based on the analysis of data obtained from experiments, it is discovered that the bimodal Gaussian density function is well fitted to the NMR T_2 spectrum and can be applied to the fitting of NMR T_2 spectrum. Furthermore, in the analysis of physical property parameters of reservoir and each individual parameter of bimodal Gaussian density function, it is indicated that the porosity and permeability are best correlated with them. Therefore, the empirical formula between porosity, permeability and the parameters of bimodal Gaussian density function were established to construct pseudo NMR T_2 spectrum by using bimodal Gaussian density function and it has been verified by the comparison between the result and experimental data. In order to achieve the more accurate and wide application of this method, the NMR T_2 spectrum versus proportion frequency curve was transformed into the pore radius R versus porosity component P_p curve before the fitting. When used in exploration field, the method can construct continuous pore size distribution curve which can be used for the quantitative evaluation of tight sandstone reservoir's quality. Compared with the method of NMR logging, this technique can be applied to the evaluation of pore structure of tight sandstone reservoirs more widely.

1. Introduction

The tight sandstone reservoir is defined as a reservoir with a porosity of less than 12% and a permeability of less than 1 mD (Kazemi, 1982; Guan, 1995; Surdam et al., 1997; Pang et al., 2003). The tight sandstone reservoir, known as an important supplement to the conventional oil and gas resources in the world today, has been attracting the attention of the world's giant oilfield enterprises, experts and scholars from its exploration to recent (Law and Curtis, 2002; Taylor et al., 2010). The tight sandstone reservoir is characterized by a low porosity, low permeability, strong heterogeneity, high water saturation, low resistivity, and complex gas-water relationship (Zeng and Li, 1994; Zou, 2012). Because of these characteristics of tight sandstone reservoir, the evaluation of it is confronted with many problems, and it is difficult for the conventional method to judge the pore structure of the

reservoir effectively (Kukul et al., 1983; Amiri et al., 2012).

The research on pore structure focuses on pore and throat, which include the size, distribution, morphology of pores and their connectivity and is of pivotal importance to evaluate the quality of reservoirs (Patrick et al., 2004). The current method is to conduct laboratory experiments on core samples, including thin section identification, scanning electron microscopy (SEM), X-ray computer tomography (CT), incremental pressure mercury injection (IPMI), nuclear magnetic resonance (NMR) and other methods. The pore structure can be characterized by these methods, such that thin section identification and SEM can give a direction observation to the two-dimensional image and size of pore and throat (Taylor et al., 2010; Golab et al., 2010; Bera et al., 2011; Shao et al., 2017); CT scanning can provide size and three-dimensional network of pores (Golab et al., 2010; Bera et al., 2011); IPMI reflects the size of pore and throat and their connectivity

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(Pittman, 1992; Yang et al., 2006; Comisky et al., 2007; Yu and Huang, 2009; Ziarani and Aguilera, 2012; Buryakovsky et al., 2012); NMR experiment mainly reflects the size and distribution of pores (Kleinberg, 1999; Ren et al., 2002), which carrying substantial weight in the evaluation of pore structure. However, these methods take long and costly and need a large number of core samples, making it impractical to conduct accurate description in the whole well interval (Shao et al., 2017; Eslami et al., 2013). The NMR log, which can be employed in the pore size evaluation in the whole well interval, can only be used in very few wells due to its high technical costs and other limitation which makes its application in all well difficult (Xiao et al., 2016; Kenyon, 1997; Prammer, 2004; Volokitin et al., 2001; Coates et al., 1999). Therefore, a method which is simpler and easier to be propagated is in desperate need to conduct pore structure evaluation continually.

For the sake of this, predecessors tried miscellaneous methods, of which the fitting of mercury injection capillary pressure (MICP) curve was frequently used (Xiao et al., 2016). The method aims to establish artificially fitted MICP curve, thereby conducting pore structure evaluation in single well continually. The method can be divided into two categories. One is that many scholars use the T_2 spectrum of NMR logging to convert them into pseudo MICP curve by the comparison of experimental data and mathematical transformation (Eslami et al., 2013; Volokitin et al., 2001; Glorioso et al., 2003; Hamada, 2009; Ouzzane et al., 2006; Arogun and Nwosu, 2011; Altunbay et al., 2001; Green et al., 2008; Looyestijn, 2001; Liu et al., 2003; He et al., 2005), including the transformation using a single scaling factor and a residual error analysis technique for the analysis of average saturation (Volokitin et al., 2001); the similar comparison method (Liu et al., 2003); and the segmentation power series method based on non-linear relation (He et al., 2005). These methods provided solutions to the issues from different perspectives. Although these were continuously ameliorated, the drawbacks of the methods have been still obvious. First of all, the prerequisite of this method is that it needs reliable NMR logging data, thus showing greater limitation that it can't be employed in the wells without NMR logging data. Secondly, the T_2 spectrum shape of NMR logging, which is affected by the properties and contents of fluids in the formation, can't reflect true pore structure in strata containing hydrocarbons, thereby making converted (MICP) curves insignificant (Volokitin et al., 2001; Ouzzane et al., 2006). The other is that some experts and scholars believed that the mercury injection pressure or mercury injection saturation for each measured points on the MICP curve is closely linked to physical property parameters of the reservoir. To make a fitting of them and to clarify the mathematical relationship and then pseudo MICP curve can be artificially fitted (Xiao et al., 2016; Liu et al., 2008; Yan et al., 2002). Zhong (1997), Liao et al. (2000), and Liu et al. (2008) established the empirical formula among pore, permeability and capillary pressure curve by using a large number of experimental data in different regions, and then fitted into the MICP curve; based on the principle and method of fractal geometry, Yan et al. (2002) used conventional physical property data to invert capillary pressure curve; Xiao et al. (2016) established an empirical formula between mercury injection saturation and porosity, and permeability after the classification of reservoir type by using J function. This method is more flexible than before, and can be widely used. However, the method has its shortcomings as follows; the artificial fitting of the curve is affected by mercury injection experiments, and the data points are constant and rare, thereby obtaining blunt curve shape which can't fully display the pore size distribution and pore structure. Moreover, high pressure fluid is used in the mercury injection experiment, which can easily damage the pore structure of the tight sandstone, resulting in the deviation of the result (Clarkson et al., 2013). In order to overcome shortcomings of above methods, based on the centrifugal core NMR experimental data of 12 core samples (Table 1) in the East China Sea Basin, and combined with porosity and permeability by formula fitting, the pore size distribution curve was established by using bimodal Gaussian density function. It is proved that the method is accurate and

Table 1

Physical properties of 12 core samples of X3-X6 well in East China Sea Basin.

Zone	Sample number	Depth(m)	Porosity (%)	Permeability (mD)
X3	1	3600.72	10.84	11.88
X3	2	3600.72	10.72	7.48
X4	3	3961.62	5.13	0.17
X4	4	3980.48	8.61	0.96
X4	5	3980.48	9.41	1.03
X6	6	4322.24	4.32	0.18
X3	7	3742.45	12.64	28.86
X3	8	3742.45	12.84	23.49
X3	9	3792.35	12.69	54.82
X3	10	3126.45	15.82	14.29
X3	11	3132.47	16.08	5.37
X3	12	3132.47	16.36	6.56

reliable, and can obtain continuous pore size distribution curve which can be used for quantitative evaluation of tight sandstone reservoir, thereby making up for the drawbacks of evaluation of tight sandstone reservoir.

2. Theory and methodology

Pore size distribution, which is associated with all the physical properties of the rock, is the core issue of reservoir characteristics of rocks (Archie, 1950). A number of models have been proposed to describe and simulate it, including a variety of density functions to simulate the pore size distribution and associated petrophysical parameters. For example, Hidajat et al. (2002) used the three-peak Weibull distribution to fit the NMR T_2 spectrum. Similarly, Genty et al. (2007) used three or more Gaussian components to fit the NMR T_2 spectrum. In fact, the Gaussian density function is often used to simulate pore size and particle size distribution in publicly reported literature (Spencer, 1963; Nimmo, 2004). Based on the previous study and combined with the experimental data, it is considered that the bimodal Gaussian density function model is the most suitable for the fitting of NMR T_2 spectrum of the tight sandstone (Fig. 1). On the basis of this, The NMR T_2 spectrum is assumed to be a superposed curve of two bimodal Gaussian density functions with six variables of T_2 .

The formula for the bimodal Gaussian density function curve (which is referred to Xu and Torres-Verdin, 2013) is as follows:

$$P(\log T_2, w_1, \log u_1, \log \sigma_1; w_2, \log u_2, \log \sigma_2)$$

$$P = w_1 * \frac{1}{\sqrt{2\pi} * \log \sigma_1} * e^{-\frac{(\log T_2 - \log u_1)^2}{2 * (\log \sigma_1)^2}} + w_2 * \frac{1}{\sqrt{2\pi} * \log \sigma_2} * e^{-\frac{(\log T_2 - \log u_2)^2}{2 * (\log \sigma_2)^2}} \quad (1)$$

$$w_1 + w_2 = S, \quad \text{and} \quad 0 < w_1, w_2 < S \quad (2)$$

Where T_2 is the transverse relaxation time in ms; P is signal proportion, in dimensionless;

w_1 and w_2 are the weighting coefficients of a single Gaussian function, and the sum of the two is the whole function area S . w_2 represents the fraction of large pore volume, which mainly determines the flow capacity and permeability, w_1 represents the fraction of small pore volume, in which irreducible water are mainly retained in the centrifugal nuclear magnetic experiment, and it shows little contribution to the fluid flow.

$\log u_1$ and $\log u_2$ are the logarithmic radius averages of smaller pore volume and larger pore volume respectively; generally larger value indicates larger pore volume which contains larger flow channel and higher conductivity; $\log \sigma_1$ and $\log \sigma_2$ are the logarithmic standard deviations of smaller pore volume and larger pore volume, respectively; it indicates the heterogeneity of the pore size; larger values indicate poor

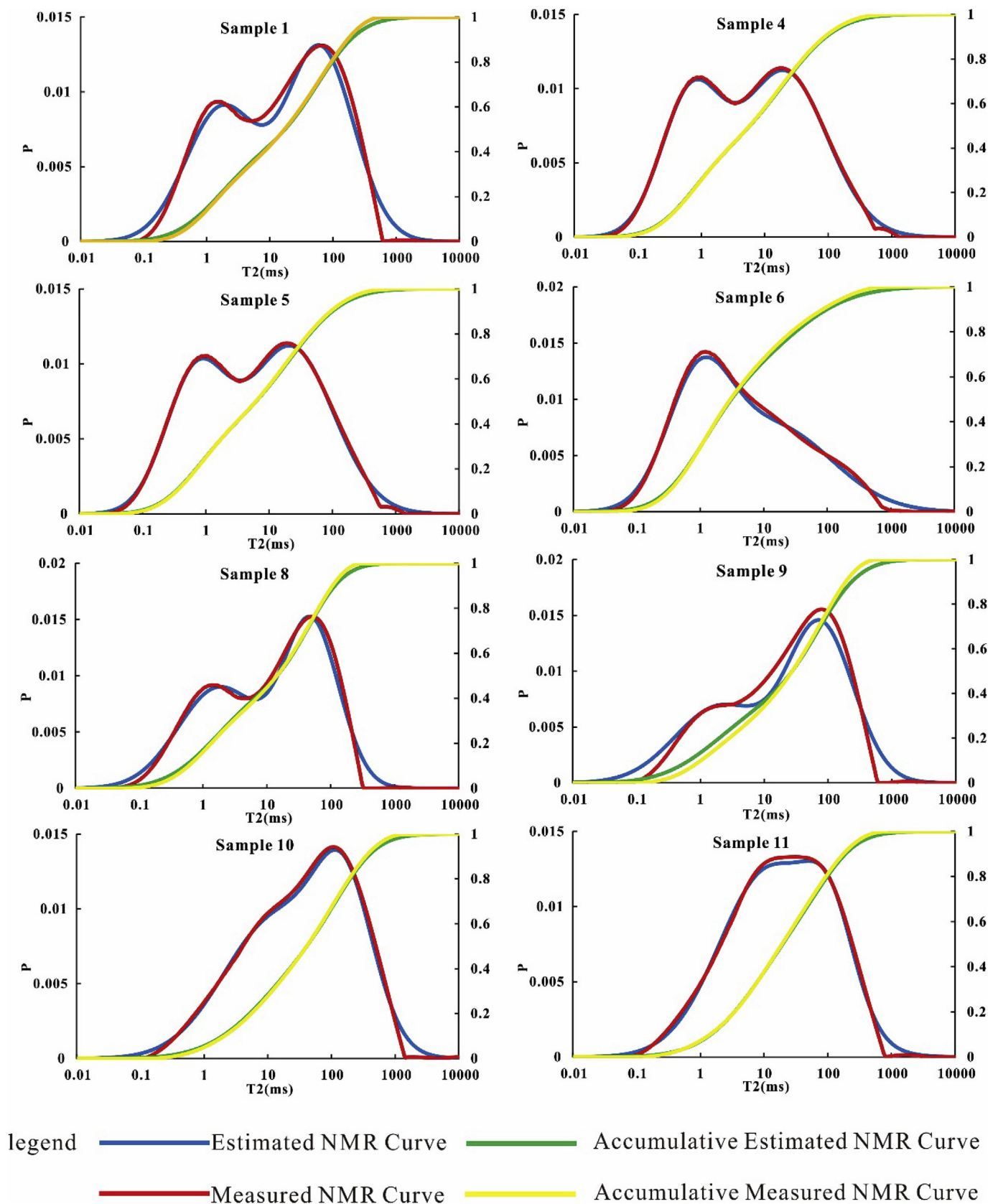


Fig. 1. Comparison between experimental NMR T_2 spectrum and fitted curves of 8 samples in East China Sea Basin.

pore sorting and higher tortuosity of pore network, resulting in a lower permeability under the same condition of pore medium radius and pore volume.

From formula (1), it can be seen that the whole bimodal Gaussian density function curve is a p -value curve about $\log T_2$ controlled by six values of w_1 , $\log u_1$, $\log \sigma_1$; w_2 , $\log u_2$, $\log \sigma_2$. The six parameters are

intimately related to pore structure properties of rocks. By the same token, parameters such as porosity and permeability are also the direct parameters that reflect the pore structure of rocks. If an experimental formula is established between physical properties (porosity, permeability) and the six parameters, then the fitted T_2 spectrum curve between $\log T_2$ and P can be established. And then it can be used to evaluate the pore size distribution of reservoir.

3. Techniques and procedures

In order to predict the NMR curve and establish the empirical formula between the basic parameters of the reservoir and the bimodal Gaussian density function, the calculation works were conducted according to the following procedures.

3.1. Procedures

The NMR data from 12 core samples of the East China Sea were used to derive the formula and the procedures are as follows:

- (1) To collect the core samples and data, and test the porosity and permeability of the core samples, and then carry out the NMR experiment under the condition of total water saturation in the samples. The measured NMR T_2 spectrum was established based on the experimental data.
- (2) According to the NMR T_2 spectrum, the parameters ($w_1, \log \mu_1, \log \sigma_1; w_2, \log \mu_2, \log \sigma_2$) of the bimodal Gaussian density function were fitted.
- (3) To establish the empirical formula between the parameters of the reservoir and the parameters of the bimodal Gaussian density function: the relationship between the parameters such as porosity and permeability, and the ratio of pore size, the mean square deviation of pore size, and peak location of pore size.
- (4) Using the empirical formula established in procedure 3, the fitted NMR T_2 spectrum curve is established by reservoir physical parameters.

Porosity and Permeability are major parameters employed in this method. The NMR T_2 spectrum curve can be fitted by data from analyzing physical properties of core samples in the coring layer, and permeability and porosity data from logging can be used in the layer without core.

3.2. Treatment and experiment of samples

A total of 12 cores were used in this test. The core was first placed in a drying oven and dried at 105 °C for 2 h. Then, the sample is placed in a high-pressure saturation device, first evacuated to -0.1 MPa for 1 h, then water-filled 20 MPa saturated device saturated for 15 h, the saturated reagent is the formation water, and the ratio is as follows (Table 2):

After entirely saturated by water, the MacroMR12-150H-I large-scale NMR imaging analyzer was used to conduct experiments on samples with total water saturation. The resonance frequency was 12.80 MHz, the magnet strength was 0.301 T, the coil diameter was 25 mm, the magnet temperature was 32 °C; the time of echo (TE) was

0.15 ms, echo number (NECH) 16000. The number of inversion was selected to be 200.

3.3. NMR T_2 spectrum fitting

A total of 12 samples from the East China Sea Basin were used to conduct nuclear magnetic experiment, and 8 samples of them were selected for curve fitting, however, the remaining four can be used for verification of reliability. At the completion of step 1, we obtained the saturated NMR T_2 spectrum which was employed in the fitting by using the bimodal Gaussian density function in *Origin* software. And the mean-square error is used as the cost function. When the mean square error is less than 5×10^{-7} , the fitting is considered a success. It is found that the fitting function has high fitting rate with the measured function (Fig. 1), with R^2 generally being more than 0.98. It indicates that the use of bimodal Gaussian density function can accurately fit the NMR T_2 spectrum curve and its attribute characteristics.

By extracting each parameters in the bimodal Gaussian density function ($w_1, \log \mu_1, \log \sigma_1; w_2, \log \mu_2, \log \sigma_2$) (Table 3), and parameters related to pore structure of reservoir were fitted, ultimately it is found that the parameters are intimately related with porosity and permeability. In order to demonstrate their correlation with porosity and permeability more clearly, the data points of each parameter of bimodal Gaussian density function and of porosity and permeability were casted in three dimensional spaces (Fig. 2). It is concluded that the parameters of the NMR curve are closely related to porosity and permeability, and the three can form a good linear relationship. The two parameters are further bimodal fitted to obtain the fitting formula in Table 4, and the fitting degree of each parameter is higher than 0.8. Among them, the curve inversion takes 200 points, the abscissa T_2 ranges from 0.01 ms to 10000 ms, $\log T_2$ range from $-2 \sim 4$, the interval between the points $\Delta \log T_2$ approximate to 0.0302; ordinate P (signal proportion) total 1. According to the formula (3), area S equals to 0.0302.

$$S = \int_{-\infty}^{\infty} P d \log T_2 \quad (3)$$

Combined with Fig. 2 and Table 4, it can be seen that the porosity, permeability have a good correlation between the parameters. By using the formula in Table 4, parameters of NMR T_2 curve can be obtained by combining porosity and permeability, thereby establishing more accurate fitted NMR T_2 curve.

For the remaining four samples, they were used to verify the reliability of the formula in Table 2. The four samples contain a porosity of 5.13%–16.36%, and a permeability of 0.17 mD - 28.86 mD. The distribution of porosity and permeability is defined as a tight sandstone reservoir and can therefore be used to verify the reliability of the method.

3.4. Reliability verifications

According to the formula in Table 4, the fitted NMR T_2 spectrum parameters of the rest four samples were obtained (Table 5), by which the fitted NMR T_2 spectrum curve was established. Then, we compared it with the measured curve (Fig. 3). It is illustrated in Fig. 3 that the fitting of NMR T_2 spectrum is in good agreement with measured curve, which can reflect the characteristics and properties of the measured curve, showing good results. Further, the R^2 values of the two range from 0.938 to 0.996, which reflects that the two are extremely close, with small deviation. Fig. 4 shows the convergence diagram of data points from experimental method and fitted curve method of the remaining four samples. The data corresponding to the X-axis is the signal proportion obtained from the NMR test, and the Y-axis corresponds to the values of signal proportion calculated by using formula in Table 4. The red line is a diagonal, which is used to indicate the deviation between the formula fit and the measured value. It can be seen that the

Table 2
Formulation of formation water (1 L).

Reagent	Quality (mg)
NaHCO ₃	1020
Na ₂ SO ₄	1500
CaCl ₂	150
NaCl	7330

Table 3
Pseudo-NMR pore size distribution curve (bimodal Gaussian density function) of the parameters.

Well name	Zone	Sample number	Depth(m)	φ (%)	K (mD)	$\log u_1$	$\log \sigma_1$	W_1	$\log u_2$	$\log \sigma_2$	W_2	R^2
Well A	X3	1	3600.72	10.84	11.88	0.243	0.577	0.013	1.798	0.534	0.017	0.988
	X4	4	3980.48	8.61	0.96	-0.162	0.476	0.011	1.306	0.678	0.019	0.999
		5	3980.48	9.41	1.03	-0.153	0.489	0.011	1.340	0.676	0.019	0.999
	X6	6	4322.24	4.32	0.18	-0.028	0.502	0.013	1.156	0.916	0.017	0.997
Well B	X3	8	3742.45	12.84	23.49	0.254	0.643	0.014	1.700	0.432	0.016	0.989
		9	3792.35	12.69	54.82	0.316	0.706	0.012	1.885	0.515	0.018	0.985
Well C	X3	10	3126.45	15.82	14.29	1.144	0.829	0.020	2.215	0.454	0.011	0.997
		11	3132.47	16.08	5.37	1.090	0.785	0.025	2.053	0.384	0.006	0.997

predicted and measured values are converged near the diagonal.
It can be seen from Figs. 3 and 4 that the method of fitting NMR T_2 spectrum curve with porosity and permeability is accurate and reliable. Once the method is applied to field operation, it is possible to establish a continuous stratigraphic fitting of the NMR T_2 spectrum data by employing accurately predicted permeability and porosity data of reservoir. Therefore, it can give a quantitative evaluation of the pore size distribution of tight sandstone reservoir.

4. Case analysis

The method is applied to the tight sandstone reservoir of East China Sea Basin for further verification. Fig. 5 is an example of D well. The distribution of the NMR pore size calculated by the above-mentioned method (bimodal Gaussian function fitting) is shown and the pore size distribution of the tight sandstone reservoir is quantitatively estimated. There are no NMR logging data in this well, but there are six depth

Table 4
Fitted formulas and fitting degree between each parameter of pseudo NMR pore distribution curve (bimodal Gaussian density function) parameters and physical properties (permeability, porosity).

Parameter	Formulae for predicting parameter	Correlation coefficient (R^2)
$\log u_2$	$\log u_2 = 0.78 + 0.047\log_{10}(K) + 0.078\varphi$	$R^2 = 0.9$
$\log u_1$	$\log u_1 = -1.217 - 0.319\log_{10}(K) + 0.157\varphi$	$R^2 = 0.827$
$\log \sigma_2$	$\log \sigma_2 = 0.902 - 0.027\log_{10}(K) - 0.03\varphi$	$R^2 = 0.801$
$\log \sigma_1$	$\log \sigma_1 = 0.272 - 0.022\log_{10}(K) + 0.032\varphi$	$R^2 = 0.812$
W_2	$W_2 = [0.207 + 0.09\log_{10}(K) - 0.01\varphi] / \varphi$	$R^2 = 0.81$
W_1	$W_1 = 0.0302 - W_2$	

points that have been sampled and subjected to NMR experiments, the formula in Table 4 can be used to fit the NMR pore size distribution and to compare it with the measured data at the sampling point, thereby verifying the result of the method. In Fig. 5, the first track shows that

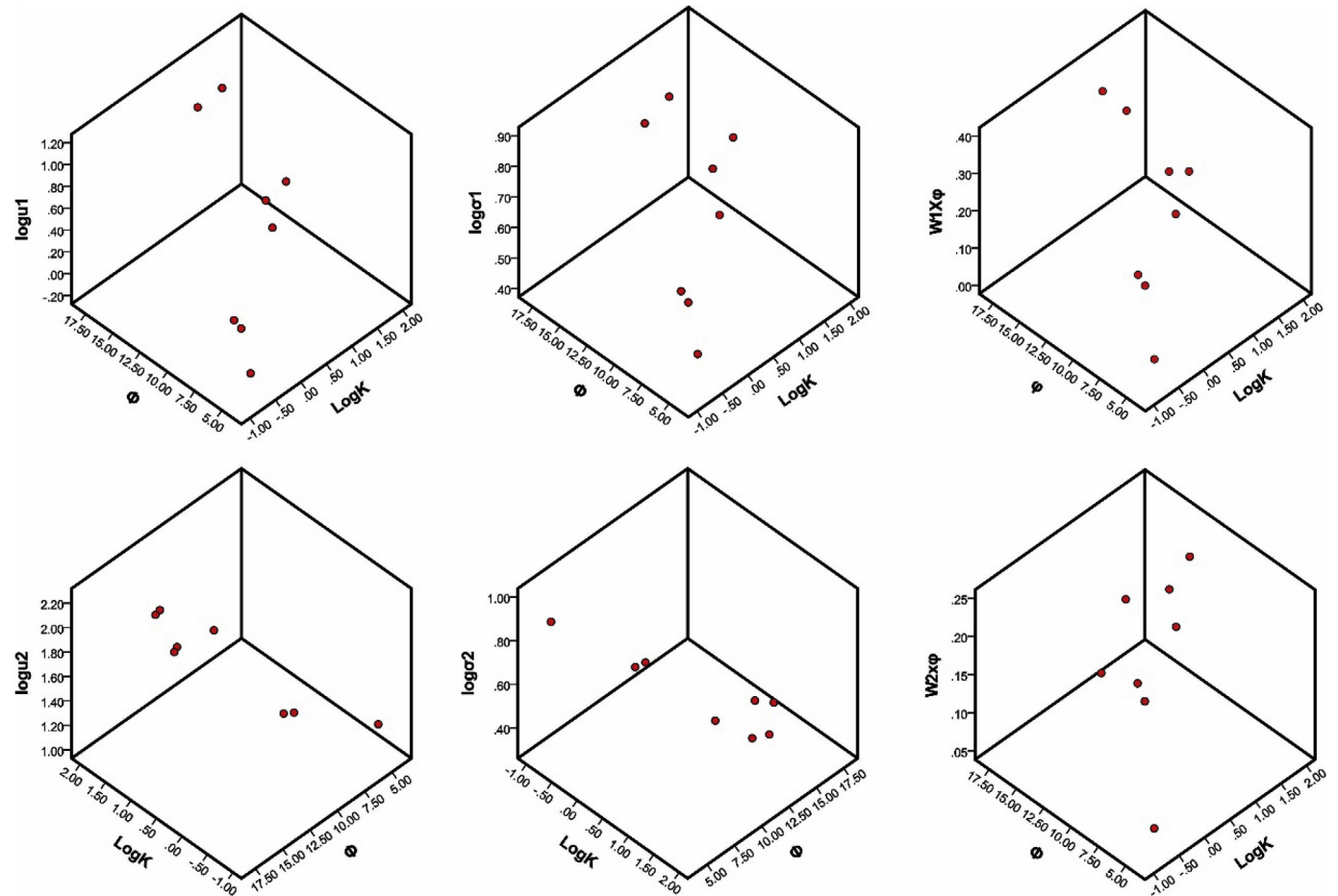


Fig. 2. Pseudo-NMR pore distribution curve (bimodal Gaussian density function) parameters and porosity, permeability 3D cast point map.

Table 5
Each parameter of pseudo-NMR pore distribution curve (bimodal Gaussian density function) of parameters for verification.

Well name	Zone	Sample number	Depth (m)	Φ (%)	K (m)	$\log u_1$	$\log \sigma_1$	W_1	$\log u_2$	$\log \sigma_2$	W_2	R^2
Well A	X3	2	3600.72	10.72	7.48	0.187	0.596	0.014	1.657	0.557	0.017	0.971
Well A	X4	3	3961.62	5.13	0.17	-0.162	0.453	0.013	1.143	0.769	0.017	0.940
Well B	X3	7	3742.45	12.64	28.86	0.302	0.644	0.013	1.835	0.483	0.017	0.938
Well C	X3	12	3132.47	16.36	6.56	1.091	0.778	0.024	2.094	0.389	0.006	0.996

the stratum is located in the X3 segment. The second track is the depth in meters. And the third track shows that the strata are mainly composed of light gray sandstone. The fourth track shows the natural gamma curve (GR), phase shift resistivity 40 inch spacing (P40H) and deep lateral resistivity (RD), all curves indicate the effective formation. The fifth track is the acoustic time difference (AC), density (DEN), and compensated neutron log (CNCF) used to estimate porosity. The sixth and seventh tracks are predicted porosity and permeability obtained by the conventional method with the practical core experimental data. It can be seen that the predicted values fit well with the experimental measurements and that the predicted porosity and permeability are credible. By using them to calculate NMR pore size distribution is reliable. The eighth track is the predicted NMR pore size distribution (PPSD). The ninth track is the comparison between the predicted NMR pore size distribution and the core measurement data. For the coring depth interval with NMR experiments, the green curve represents the predicted NMR curve (PPSD) and the red curve represents measured NMR curve (MPSD). From the comparison, it is indicated that the two are almost consistent. In the tenth track, in contrast, we compared the measured mean NMR radius (MMR from NMR) with the predicted mean NMR radius (PMR), and the measured value agrees well with the predicted value. It is proved that the fitted NMR pore size distribution

curve is in good agreement with core experimental results. The eleventh track is the mean throat radius (MMR from IPMI) measured in mercury injection experiment and the predicted mean NMR pore radius (PMR). And the difference between the two is generally 1 μm , which is caused by the fact that mercury injection mainly reflects breakthrough throat radius, while NMR reflects pore volume. It results in a lower value in the mercury injection experiment than that in NMR experiment. Meanwhile, the high pressure condition used in the mercury injection experiment can also affect the pore structure and clay minerals in tight sandstone, which may also be the origin of the differences between the two (Clarkson et al., 2013). However, although the two values are different, they show similar varying trend in the curve, indicating that the pseudo NMR curve can fully reflect the reservoir pore structure and quality of real strata. It proved the reliability of the fitted curve and feasibility of the method as well.

5. Discussion

According to the formula in Table 4, it is known to us that two parameters affecting the fitting curve are porosity and permeability. However, after accurately obtaining the permeability and porosity by logging, the error of the curve is mainly due to the differences between

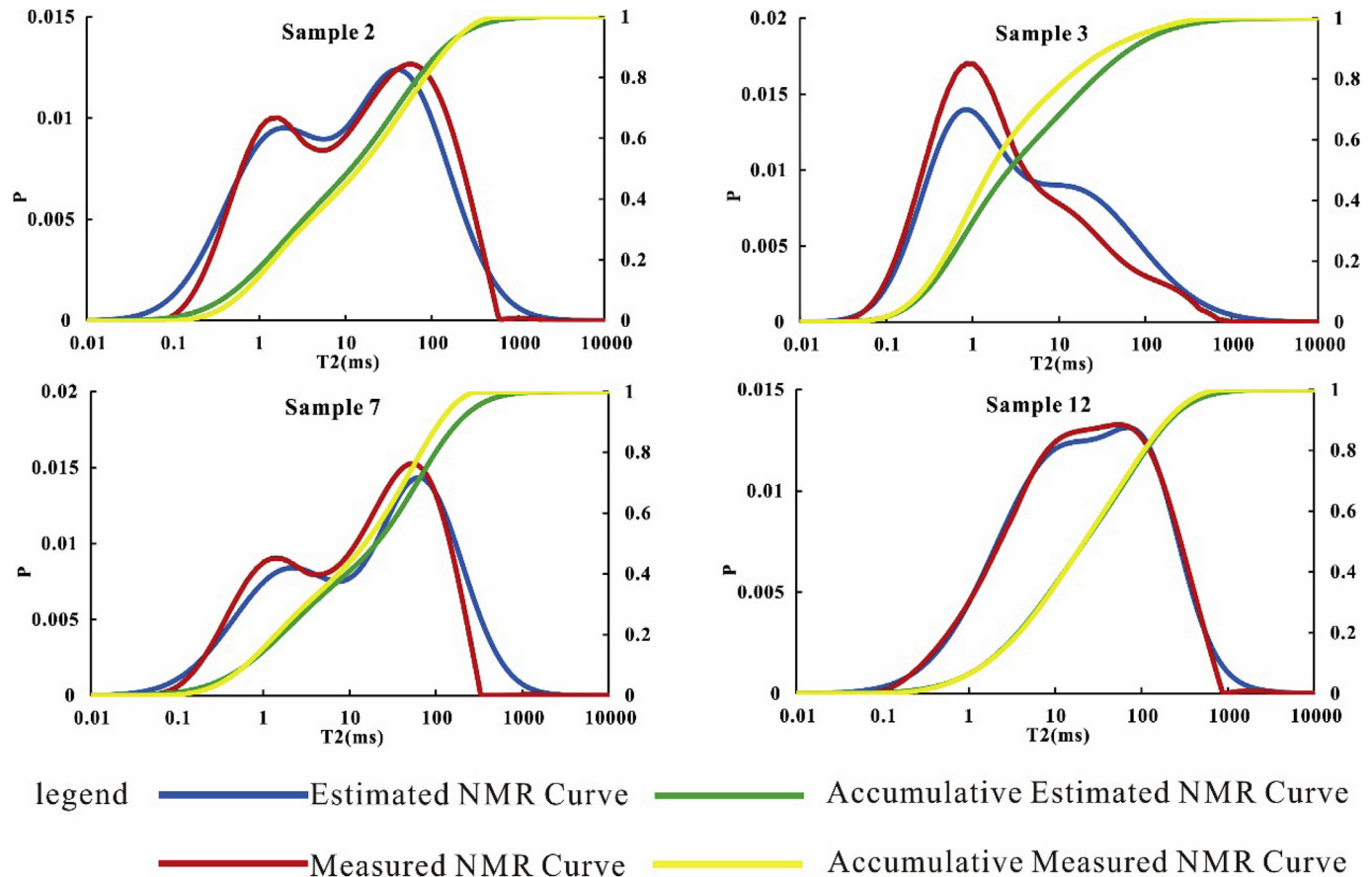


Fig. 3. Comparison of experimental results of NMR test curve and fitting curve of 4 samples in East China Sea Basin.

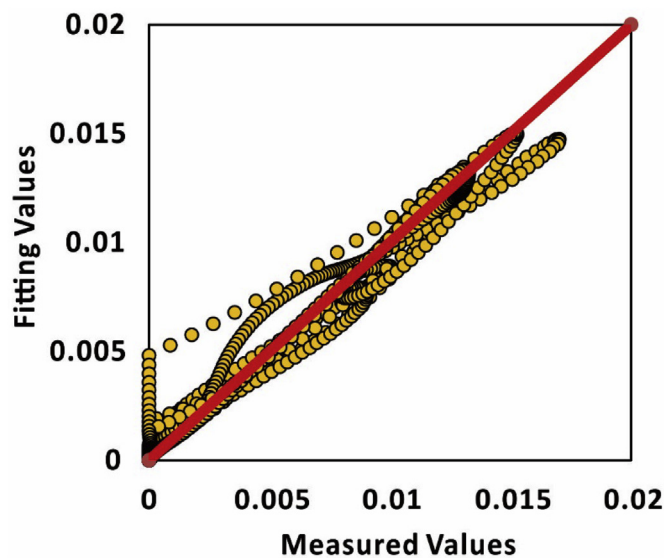


Fig. 4. Comparison of experimental results of nuclear magnetic test and fitting curve of 4 samples in East China Sea.

the measured curve and the bimodal Gaussian density function. For instance, the curve of most measured data showed a cliff-shaped decline in the right, however, due to the characteristics of bimodal Gaussian density function, the curve gradually slowed down in the right, resulting in greater differences in the right between the two. The fitting value is generally greater than the measured value. Meanwhile, due to equivalent area of the two curves, the slope of the values in the right of fitted curve was smaller and the curve was dispersive, generally resulting in a lower value of fitted value that measured value in the peak or ebb between two peaks. In Fig. 4, it can be seen that in the low value intervals (the left and right ends of the curve), the yellow dots were generally above the diagonal, indicating that fitted value was greater than measured value; in the high value intervals (the peak or ebb intervals between peaks), the yellow dots were generally below the diagonal, indicating that fitted value was smaller than measured value. It is indicated by calculation that the error of fitting curve for the samples ranges from 2.2% to 18.7%, with an average of 8.8%. The error of the larger modal in the right is 0.8%–9.2%, accounting for 11.4%–49.4% of the total error. The differences between this part of fitted curve and the measured data account for a larger part of the error, but we can see that the overall error is still minor. From Figs. 1 and 3, the fitted curve

accurately reflects the trend of the measured curve and a variety of parameters. And the error is in an acceptable range, thus verifying the accuracy and reliability of the fitting curve. And the method can be applicable.

In the above experiment, the experimental data of the samples are carried out under the same experimental conditions, so there is no need to make a correction between the curves. In the curve, the $NMR\ T_2$ spectrum was taken as the abscissa, and the signal proportion P was taken as the ordinate. However, when the samples in different regions under different experimental conditions are compared with each other, the $NMR\ T_2$ spectrum of the abscissa may also be shifted due to the different petrophysical properties, mineral composition, changes in testing instrument and inversion formulas. In this case, it might bring out a large deviation if the measured curves under different experimental conditions were fitted by using the formula in Table 4. Therefore, in order to make the convenient fitting of samples under different conditions and a contrastive observation of pore size distribution more intuitively, the $NMR\ T_2$ spectrum was converted into the pore radius R for convenient comparison between different samples by using previously mentioned empirical methods (including similar comparison method and minimum average saturation error method).

Variations in experimental conditions will not only affect T_2 spectrum in the abscissa, but also lead to the deviation of P in the ordinate. Under different experimental conditions, different numbers of inversion points were selected. When the number of points is small, the points are dispersed distribution, and the P is relatively concentrated. The ordinate signal proportion P is larger and the curve becomes higher. When the number of points is larger, the data points are in denser distribution, and the P is scattered, the ordinate P becomes smaller. Therefore, the curve became shorter. In this case, the shape of various data curve differs greatly and can't be fitted. Firstly, each curve was integrated to unify curve area (S), the P is converted into the one when the curve area was 1 (Equation (4)). And then the fitting is carried out. Meanwhile, taking the area P as the ordinate is not conducive to the comparison between samples with different porosities. After the normalization of curve area, the ordinate P was multiplied with porosity, and then it was converted into pore radius component (P_p). And it can be convenient in practical comparison.

$$S = \int_{-\infty}^{\infty} P d\log R = 1 \tag{4}$$

Fig. 6a shows the characteristics of the sample curve when different inversion points are selected. Fig. 6a illustrated the curve comparison of two samples, the red line represents that 200 points were selected and

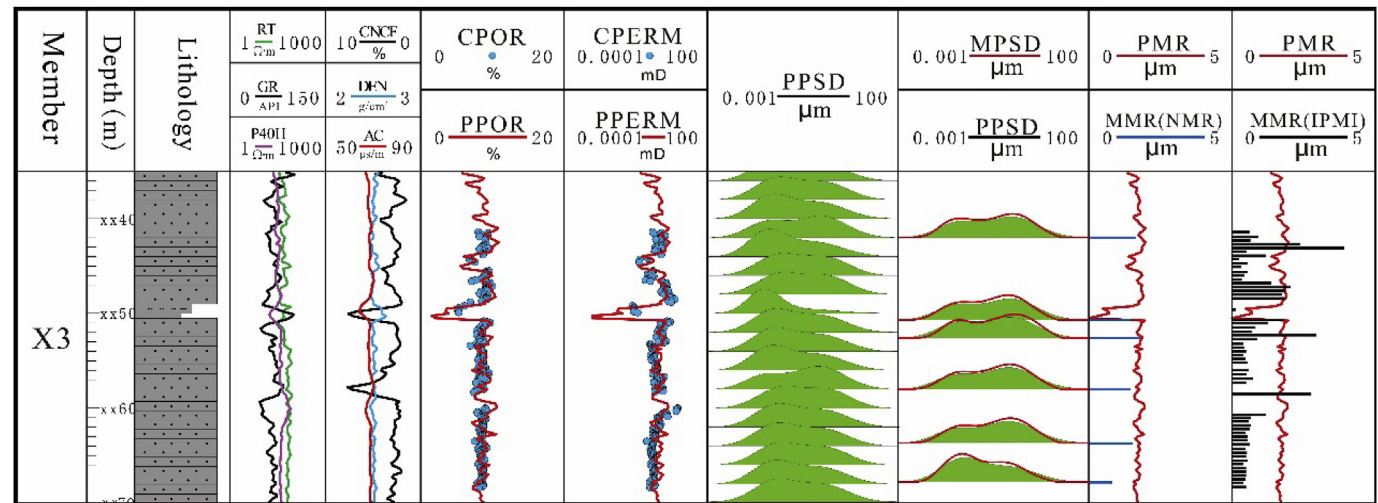


Fig. 5. Comparison of magnetic and magnetic curves obtained by two methods.

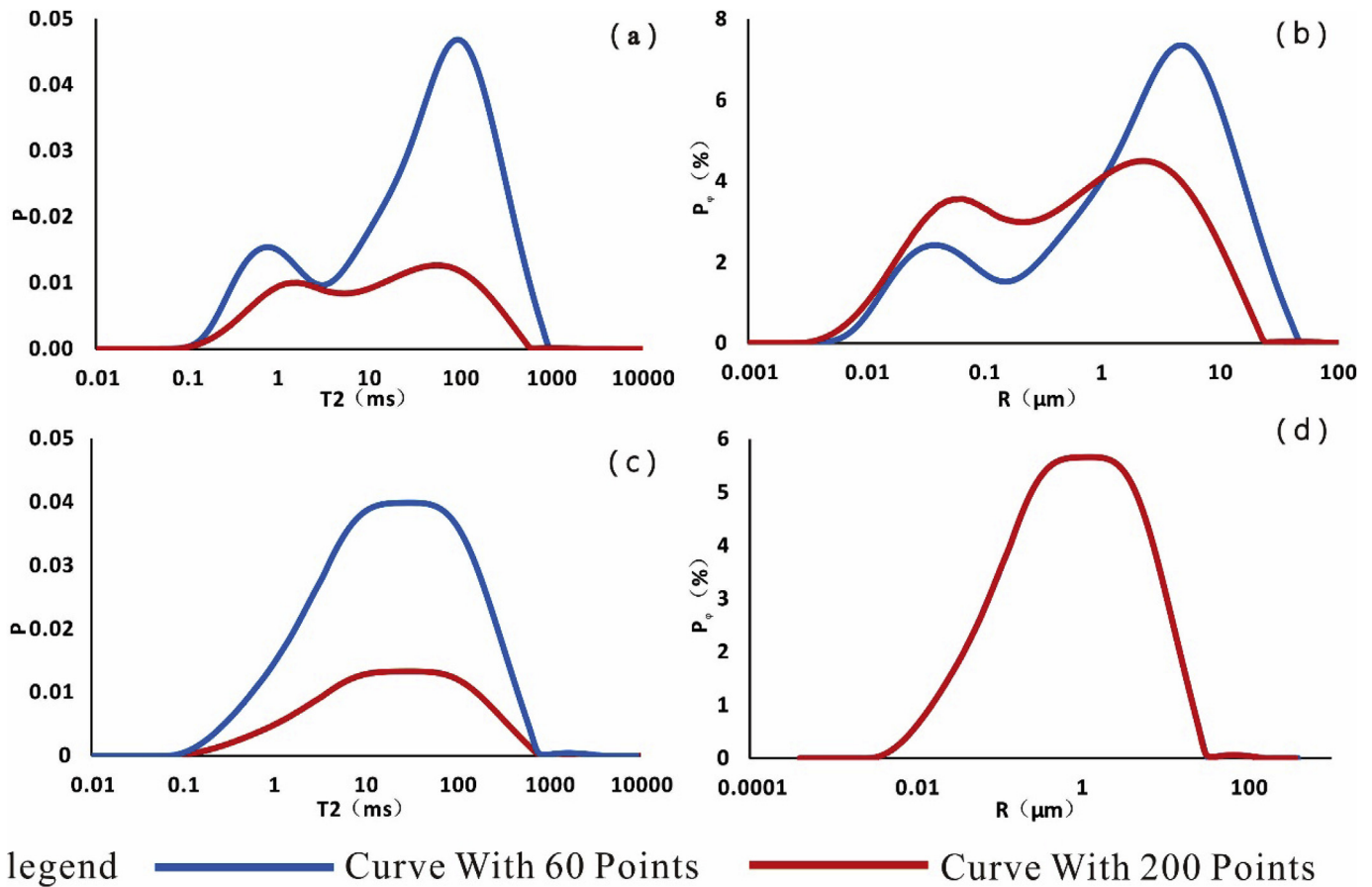


Fig. 6. Comparison between original NMR T2 spectrum and NMR T2 spectrum with ordinate conversion.

Table 6

Fitting formula and fitting degree between each parameter of pseudo-NMR pore distribution curve (bimodal Gaussian density function) and permeability, porosity.

Parameter	Formulae for predicting parameter	Correlation coefficient (R^2)
$\log u_2$	$\log u_2 = -0.618 + 0.047 \log_{10}(K) + 0.078 \phi$	$R^2 = 0.9$
$\log u_1$	$\log u_1 = -2.615 - 0.319 \log_{10}(K) + 0.157 \phi$	$R^2 = 0.827$
$\log \sigma_2$	$\log \sigma_2 = 0.902 - 0.027 \log_{10}(K) - 0.03 \phi$	$R^2 = 0.801$
$\log \sigma_1$	$\log \sigma_1 = 0.272 - 0.022 \log_{10}(K) + 0.032 \phi$	$R^2 = 0.812$
W_2	$W_2 = [6.85 + 2.98 \log_{10}(K) - 0.3 \phi] / \phi$	$R^2 = 0.81$
W_1	$W_1 = 1 - W_2$	

the curve is shorter; the blue line represents that 60 points were selected and the curve is higher. There is great difference between the two samples, thus making the fitting difficult. Fig. 6c illustrates the comparison of curves from the same sample but with different inversion points, and the differences between the two are large. Therefore, under the condition of conversion of abscissa T_2 into the pore radius R , we integrated the areas of the two curves and unified to define them as 1. At this moment, the sum of w_1 and w_2 is 1 due to the properties of bimodal Gaussian density function fitting curve. Meanwhile, in order to facilitate the comparison between samples with different porosities, the ordinate is multiplied by the porosity and converted into the aperture component (Fig. 6b and 6d). Therefore, when fitting the sample data under different experimental conditions, the abscissa and ordinate are first unified, that is, the NMR T_2 spectrum is converted into the pore radius R , and then the curve area is normalized by integration. Finally, for the convenient division of samples with different porosities, the function was multiplied by the value of porosity as a whole. The fitting

curve formula can be converted into the following:

$$P\phi(\log R; w_1, \log u_1, \log \sigma_1; w_2, \log u_2, \log \sigma_2)$$

$$P_\phi = w_1 * \frac{1}{\sqrt{2\pi} * \log \sigma_1} * e^{-\frac{(\log R - \log u_1)^2}{2 * (\log \sigma_1)^2}} + w_2 * \frac{1}{\sqrt{2\pi} * \log \sigma_2} * e^{-\frac{(\log R - \log u_2)^2}{2 * (\log \sigma_2)^2}} \quad (5)$$

$$w_1 + w_2 = 1, \quad \text{and} \quad 0 < w_1, w_2 < 1 \quad (6)$$

Where R represents the pore radius in μm , ϕ represents porosity of samples, in%, and the rest parameters have the same meaning as formula 1. For converted curve, the abscissa is pore radius R , the ordinate is porosity component P_ϕ . That is, the T_2 - P diagram is converted into the R - P_ϕ diagram (Fig. 6), and then the parameters of bimodal Gaussian density function are determined by the above method to establish the formula for the sample curve fitting (Table 6). After determining the curve fitting formula, the porosity and permeability data can be used to establish the fitted NMR pore size distribution curve, to predict pore size distribution, and to evaluate pore structure of reservoir.

Lithologies of the target horizon in the study area vary slightly. And the rocks in the horizon experienced similar burial history and diagenesis, thereby making the method feasible and reliable in the study area. Meanwhile, the target reservoir with a relatively simple pore structure is characterized by few micro-pores and good correlation between permeability and porosity, that it is consistent with Hu et al. (2013) and Li (2015). If the geological backgrounds in other areas are similar that in the study area, this method is likely to be applicable. However, for reservoir dominated by micro-pores, it shows complex structure and relatively poor correlation between porosity and

permeability. This method might be applicable by combining with other parameters or fitting by categories that it is our next research direction.

6. Conclusions

Evaluation of the pore size is an indispensable part of the identification and evaluation of tight sandstone reservoir, and it is difficult to evaluate the tight sandstone reservoir effectively by conventional method. The complex pore structure can be quantitatively described by using the bimodal Gaussian density function. The parameters can be used to judge petrophysical properties of rocks, and the following conclusions were made by combining with the researches on the samples in East China Sea Basin.

- (1) By analyzing the NMR curve data, the bimodal Gaussian density function is selected to fit the NMR curve, showing a high fitting degree. For eight samples among them, it is above 0.98. Therefore, the NMR T_2 spectrum curve and its attribute characteristics can be accurately fitted by using bimodal Gaussian density function.
- (2) To establish the corresponding relationship between the parameters of the bimodal Gaussian density function and physical properties (permeability, porosity), further the accurate permeability and porosity data from logging can be employed to establish NMR T_2 spectrum curve. By using this method, there are high fitting degree and small errors between fitted curve and measured curve. And reliability of the method has been verified by NMR experimental data.
- (3) Under different experimental conditions and basic conditions of NMR curve, the deviation is large, making it difficult in fitting the curve. Therefore, before fitting, the experimental data need to be pretreated, converting T_2 - P curve into R - P_ϕ curve. After this treatment, the fitting curve is more accurate and more convenient to analyze the contrast.

The method can be widely used in the evaluation of tight sandstone reservoirs. In the absence of NMR logging, the method can achieved good results on the aspects such as pore size distribution, pore structure parameters and identification of effective tight reservoirs which can't be identified by conventional methods.

Author contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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